

REPORT ON:

PLAQUE DETECTION FROM ULTRASOUND IMAGES

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ABSTRACT:

This report describes an automated algorithm for plaque detection in Intra Vascular Ultrasound images, using an adaptive border detection method for selection of Region of Interest (ROI) and detect the plaque inside ROI. Our algorithm uses feature extraction from the images and detecting the lumen and media boundaries. This narrows down the search of plaque inside the ROI.

INTRODUCTION:

Intravascular ultrasound (IVUS) is an imaging technique that is capable of providing transluminal tomographic images of coronary arteries in vivo. While coronary angiography provides only a silhouette of the coronary artery lumen, the images that IVUS show the coronary artery wall, plaque morphology. Thus the information about the coronary artery that IVUS provides can be of great use in determining coronary intervention strategies. As IVUS catheter technology has improved, the range and number of patients whom IVUS can benefit from has increased considerably. It has further been reported that the negative contrast technique and coronary injection of normal saline are useful for enhancing the diagnostic capabilities of IVUS. While the investigation and assessment of the pathological characteristics of acute coronary syndromes is quite advanced, their mechanisms are still not well understood. The high incidence of thrombus in lesions in the acute phase of myocardial infarction is perhaps one reason why there has been insufficient investigation of plaque in these patients. The purpose of this study was to investigate plaque morphology at the time of acute myocardial infarction.

STATE OF ART:

Manual lumen and media segmentation is a laborious task that suffers from inter- and intraobserver variabilities due to the high amount of noise and artefacts present on the IVUS images. To overcome these limitations, several approaches for semi-automated segmentation have been proposed in the literature. Sonka et al implemented a knowledge-based graph searching method incorporating a priori knowledge of coronary artery anatomy and a selected region of interest prior to the automatic border detection. Quite a few variations of active contour model have been investigated including the in vivo validation approach of Giannoglou et al. The active contour principles have been used to allow the extraction of the borders in three dimensions after setting an initial contour in Kovalski et al's approach. However, the contour detection fails for low contrast interface regions such as the luminal border where the blood-wall interface in most images corresponds to weak pixel intensity variation. In clinical practice the most attractive approaches are the fully automatic ones. Our method mainly includes segmenting catheter, lumen and the media, such as the segmentation based on edge contrast, the latter is shown to be an efficient feature for IVUS image analysis, in combination with the gray level distribution.

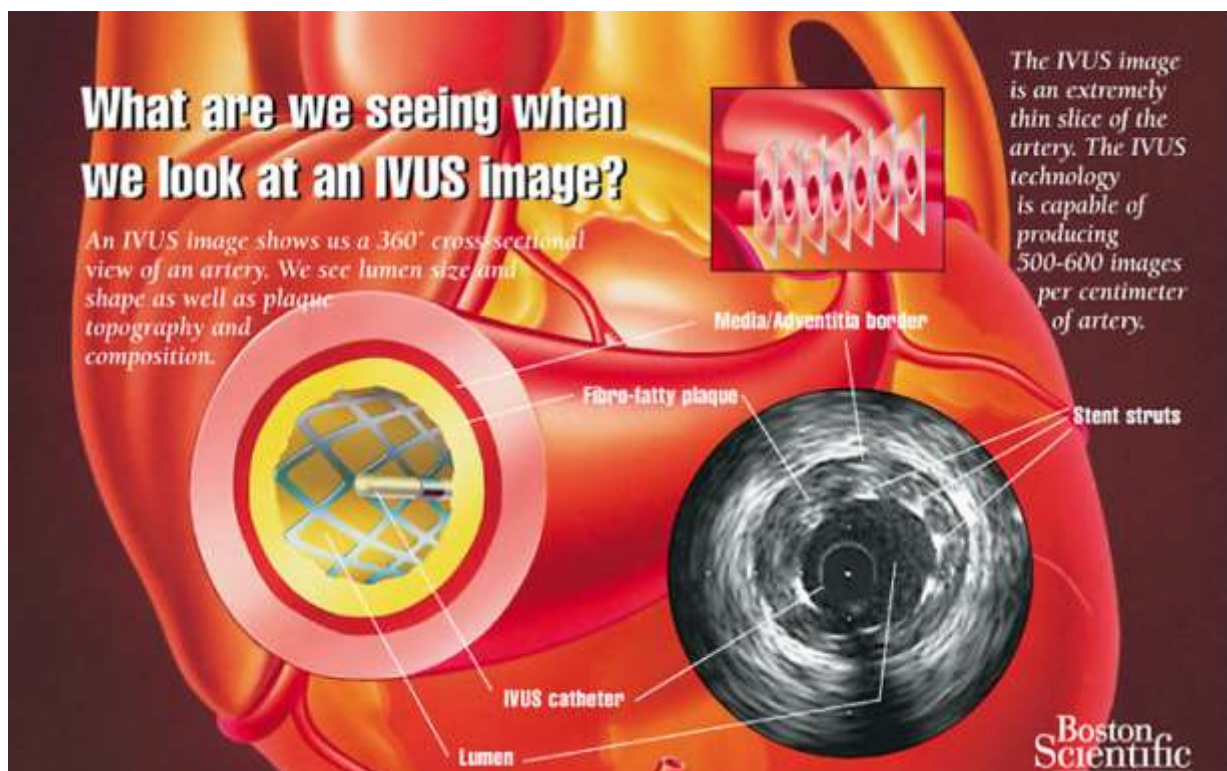
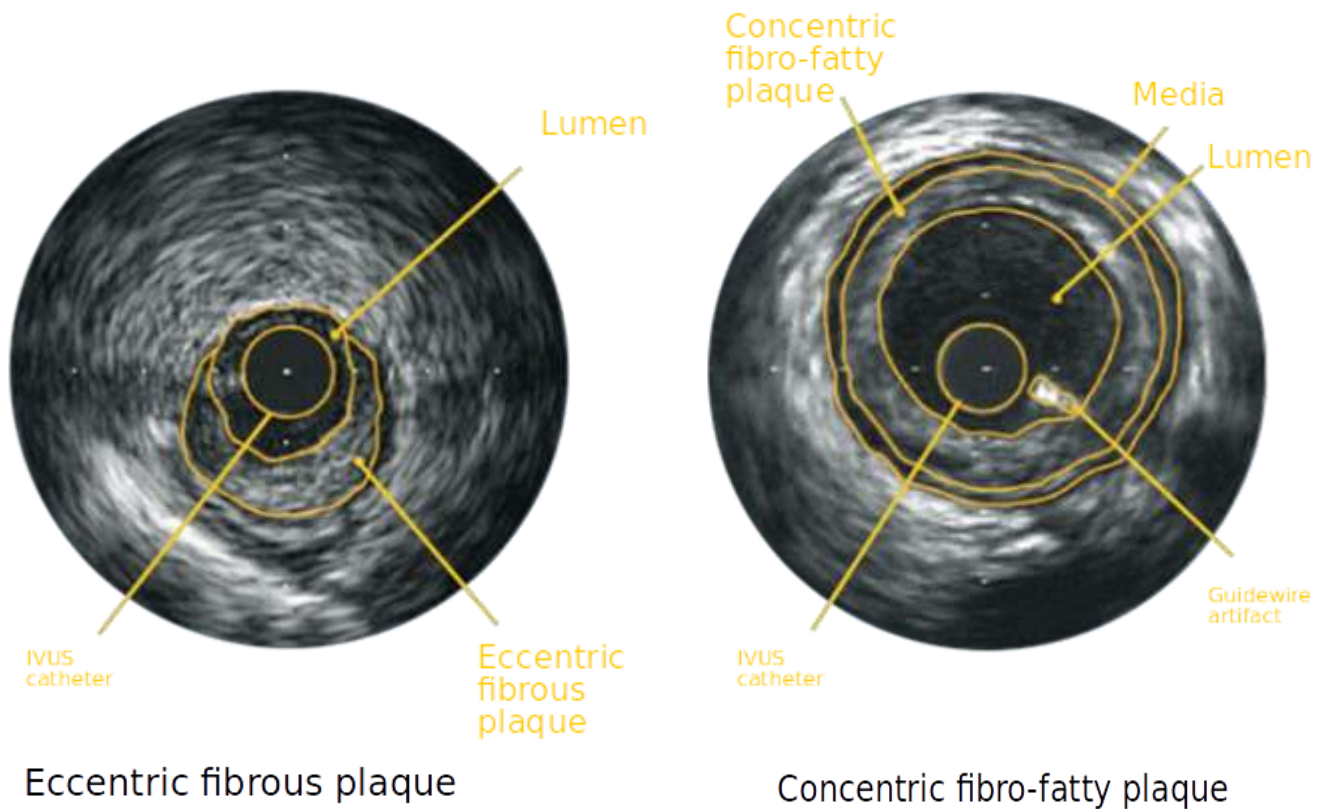


IMAGE INTERPRETATION



DATASET

Dataset A: This is a set of 50 images extracted from in vivo pullbacks of human coronary arteries, from 20 patients. The Imaging System used for the acquisition is an iLab IVUS (Boston Scientific, Fremont), equipped with a 40 MHz catheter Atlantis SR 40 Pro[5].

ALGORITHM FLOW:

Since the data set consists of plaque morphology in different shapes, we try to fit a convex shape around the lumen and then minimize our detection zone and apply gradient thresholding to find out the plaque region.

PROCEDURE:

1. Best Image size to deal with:

The Data Set is converted from 512×512 to different sizes 256×256 , 128×128 to check the amount of information lost and increase in speed of execution. 256×256 is used as it has minimal execution time and information loss. The image is quantised into 3 bits from 8-bit grey scale image. 8 levels (3 bits) indicated a better thresholding factor.

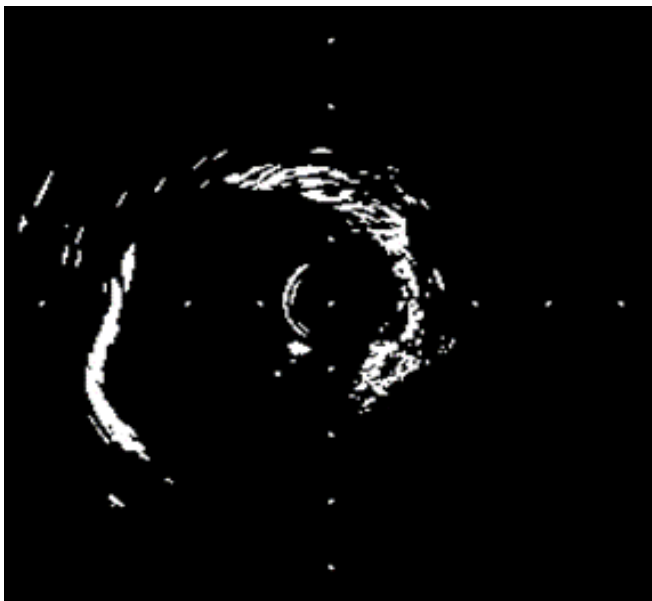


Fig 3. 256x256 (3 bits)

IMAGE INVERSION:

The image is inverted as the lumen is brightened using a threshold of 20, as the outer unwanted area in the data set remains at low intensity. The outer membrane is intensified.

IMAGE EROSION AND DILATION:

Using Media to separate the Lumen and Adventitia layers, erosion and dilation have been used with kernel size 1 and 2 respectively.

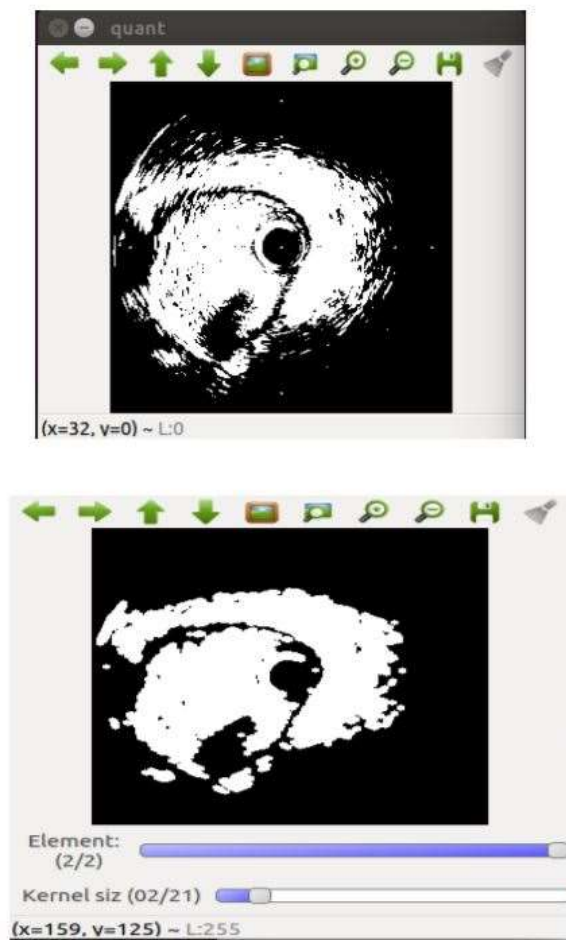


Fig 4. After Erosion + Dilation

MASKING AND FILLING:

The noise and unnecessary Blobs inside the Lumen and Adventitia are filled with a mask around the boundary layers.



Fig 5: Removing noise, inside blobs

CONVEX HULL FIT:

The convex hull of a finite point set S is the set of all convex combinations of its points. In a convex combination, each point X_i in S is assigned a weight or coefficient A_i in such a way that the coefficients are all non-negative and sum of them is one, and these weights are used to compute a weighted average of the points. For each choice of coefficients, the resulting convex combination is a point in the convex hull, and the whole convex hull can be formed by choosing coefficients in all possible ways. Expressing this as a single formula, the convex hull is the set:

$$\text{Conv}(S) = \left\{ \sum_{i=1}^{|S|} \alpha_i x_i \mid (\forall i : \alpha_i \geq 0) \wedge \sum_{i=1}^{|S|} \alpha_i = 1 \right\}.$$

Contour is fit around the lumen and a convex polygon is fit around the boundary set.

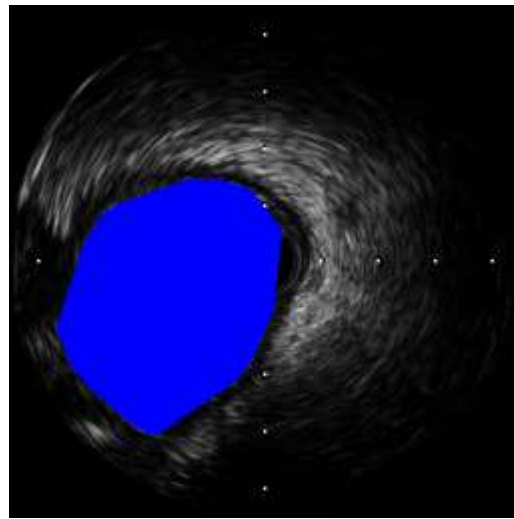


Fig 6. Convex Hull fit of the lumen area

Plaque segmentation:

The inner, finer ROI is accessed for bright intensity distribution. The bright most parts with large active contour areas are found out. Now, the contour is filled with a random colour. The whole image is traversed to find out the colour and the real values of that are mapped into a blank image of same resolution. The process of contour detection is used again on the remaining bright distributions which include plaque morphology.



Fig7. Plaque Segmentation (Intensity)

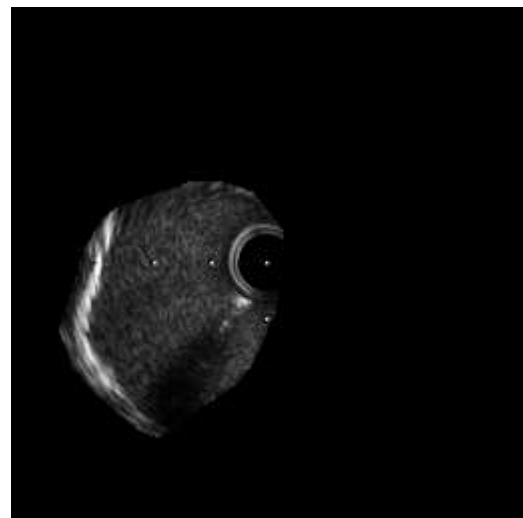


Fig8. Lumen and Plaque extraction

Results:

Image 1 and 2 have concentric plaque and Images 3 and 4 have Eccentric plaque

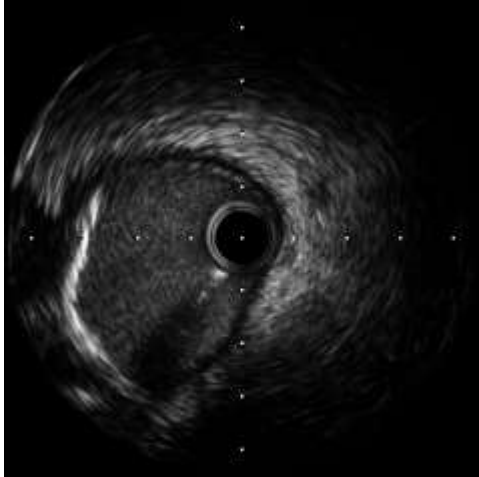
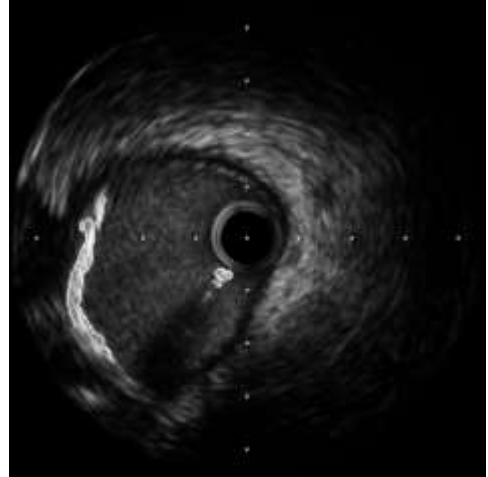


Image 1



Plaque detected in Image 1

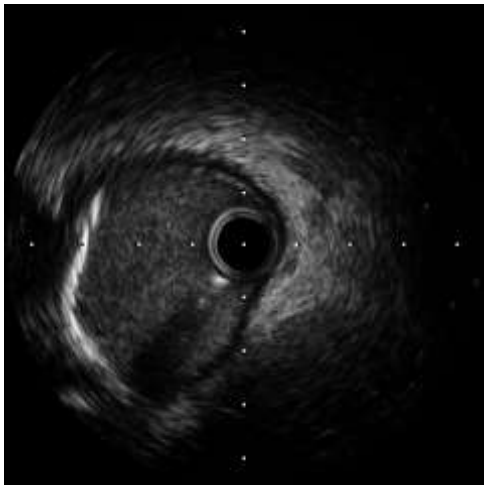
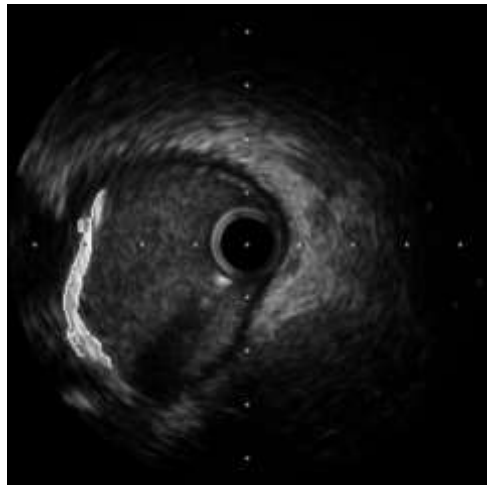


Image 2



Plaque detected in Image 2

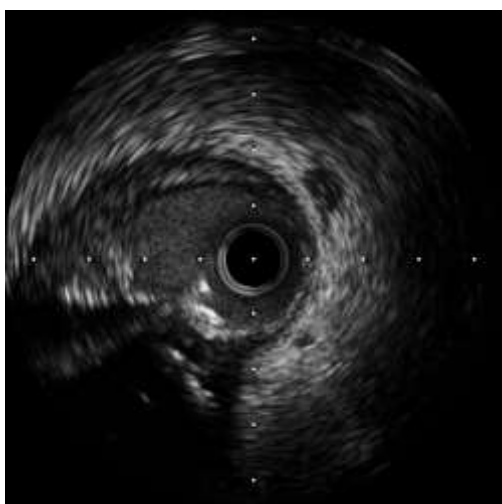
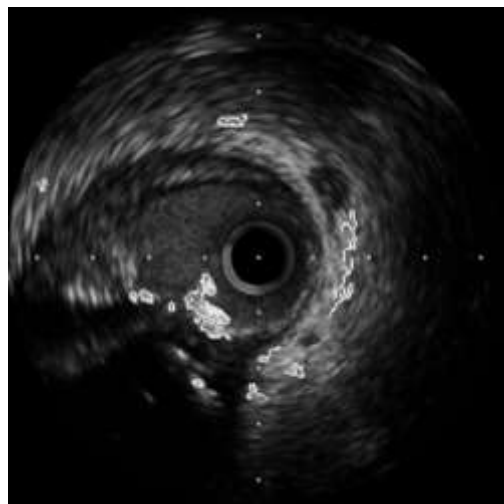


Image 3



Plaque detected in Image 3

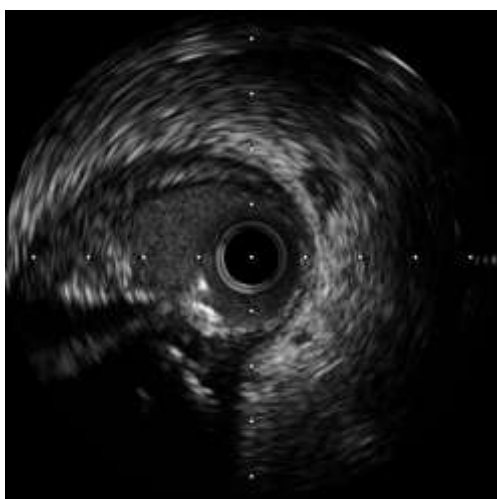
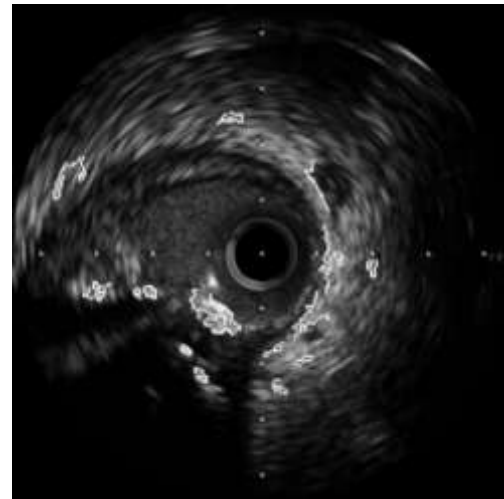


Image 4



Plaque detected in Image 4

FUTURE WORK

- Shadow region analysis in the images can be used to get the details about calcified plaque which is obstructing the ultrasound to pass through it.
- Extending this approach to random plaque distribution using contrast gradient and feature extraction.
- Complete automated algorithms can be developed using learning methodology.

REFERENCES

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